Study Report on Project of Submarine Cables

Zhuoyang Lin

Abstract

With the acceleration of urbanization and the continuous improvement of power systems, submarine cables are increasingly widespread due to their high reliability in power supply and minimal occupation of sea areas. Since July 2024, I have had the privilege to take part in the research project of State Grid Xiamen Power Supply Company "Research on Submarine Cable Simulation and Status Evaluation Technology Driven by Data and Physics", where I embarked on preliminary scientific research explorations.

After targeted field research, I have gained certain knowledge about the types, structure, laying, manufacturing, and daily operation and maintenance (O&M) of submarine cables. Based on this, I completed the research about methods for structural classification and quality improvement of submarine cable data as a team member. In order to make effective evaluations for the condition of submarine cables, a large amount of data needs to be analyzed and filtered. The premise of this work is to obtain high-quality data, while the data obtained on-site usually have the problem of disorganized classification and missing values. Therefore, various methods are needed to improve the quality of the data. In this process, by reviewing relevant literature sources, I analyzed data types, organized and classified the data based on usage, and employed a number of techniques, including several statistical methods and artificial intelligence methods, to fill in missing data values and effectively improve data quality.

Through taking the job of data's structural classification and quality improvement, I have initially developed the skills such as the summarization and organization of sources and experimental research, and formed a preliminary understanding of technical research methodologies.

1. Structural Classification of Data

As submarine cable technologies are continuously used in the construction of power systems, O&M departments have accumulated a large amount of operational data from actual engineering projects. These raw data vary in format and granularity, with issues such as inconsistent naming, complex data nesting, and excessive redundant information, therefore require systematic structural classification.

In addition, the time span, sampling frequency, and spatial distribution of the data vary, increasing the difficulty of structural processing. For example, cable operation data within the same time period might be recorded in seconds, while manual inspection records are typically recorded daily or weekly. This requires us to simultaneously consider both time alignment and fusion capabilities when performing structural classification. This approach can contribute to increasing the efficiency of operation monitoring, fault diagnosis, and O&M management.

1.1. Data Collected in This Project

The data collected in this project include cable equipment information, cable operating voltage, current, active power, temperature, partial discharge information, grounding current information, fault alarm data, environmental monitoring data, cable connection test data, periodic testing data, and various inspection records. These data have various sources, including data from online monitoring systems and intelligent sensors, as well as manual field records and O&M ledger system exports.

1.2. The Data Structures and Categories Used in Existing Cable Condition Assessment Methods

Currently, there are multiple methods for evaluating the condition of submarine cables.

Literature[1] describes a method that monitors the condition of fiber-optic composite submarine cables by detecting optical components. By analyzing frequency shifts through online monitoring, the fiber's temperature and strain data can be obtained, and then be used to assess the cable's condition.

Literature[2] introduces a cable condition assessment method based on the log-normal distribution. This method evaluates cable aging by analyzing dielectric loss and partial discharge data. The results of this research helps in timely understanding of the cable's condition during operation and in proposing appropriate maintenance plans.

Literature[3] investigates the mechanism of insulation decrease in high-voltage cables. A mathematical model for cable insulation is established, and the integration of artificial neural networks, expert systems, virtual instruments, and intelligent databases is explored for fault diagnosis and early warning. Various online detection methods for characteristic parameters are discussed, along with the intrinsic relationships between these parameters and faults, as illustrated in Figure 1.

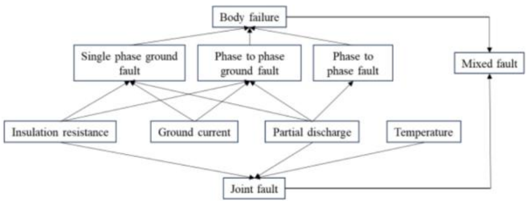

Figure 1. Relationship Between Characteristic Parameters and Cable Faults

Literature[4] begins with a discussion on the operation and maintenance (O&M) management of power cables and proposes a lean management process in accordance with the ISO 5000 international asset management standard. This process includes asset registration, condition monitoring, status assessment, risk analysis, and the formulation of O&M strategies. It then summarizes recent advances in cable condition monitoring, condition assessment, and fault data analysis. These advances include the use of new condition monitoring technologies, fault location techniques, and statistical tools to process cable fault data and predict the number of future failures. Research has also analyzed the remaining life of cable insulation using thermo-electrical aging models. Finally, it also introduced a retirement management strategy, which is based on combining condition assessment with asset criticality via risk analysis. The classification of cable data structures is shown in Figure 2.

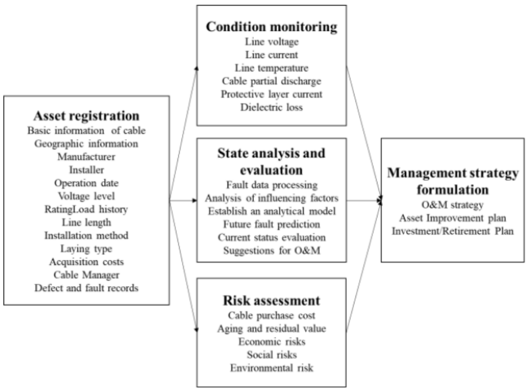

Figure 2. Classification Diagram of Various Cable Parameter

In conclusion, regardless what type of method for evaluating cable condition is used, the requirements for data input always have some common characteristics:

1. The input data must follow a unified structure.

2. The data should possess time-series properties and integrity.

3. The evaluation models require multi-dimensional feature variables, including electrical, thermal, environmental, and mechanical stress parameters.

1.3. Results of Data Structuring and Classification

According to studies of literature sources and in-depth investigation of existing types, origin, and usage scenarios of data, this report proposes a data structuring and classification method suitable for this project. According to data attributes, data related to submarine cable are structured into the following six main categories:

1. Electrical Data: Includes voltage, current, active/reactive power, frequency, phase angle, etc. These are the vital data that reflect the basic conditions of cables. They usually come from online monitoring systems or substation SCADA systems, and can be used to determine load levels and identify electrical anomalies.

2. Thermodynamics Data: Such as temperature of conductor, insulator and sheath, along with temperature of surrounding seawater. This type of data is crucial for assessing the thermal

lifespan, overload risks, and environmental conditions of cables during their operation.

3. Mechanical Status Data: Monitors physical stress variables such as cable tension, bending stress, and vibration intensity. These data are essential for evaluating cable health, especially in floating or geologically active areas.

4. Environmental Impact Data: Include seawater temperature, air temperature, etc. These data are usually supplied by third-party marine monitoring systems, and serve as background information for cable laying and maintenance strategy development.

5. Operational Status and Event Data: Include alarm logs and fault diagnosis records. These data help to trace historical anomalies and to construct fault evolution pathways.

6. Operations and Maintenance (O&M) History Data: Include inspection logs, maintenance records, construction diaries, etc. These records assist in analyzing the relationship between manual interventions and change in facility conditions.

In addition to attributes, our team also classified the data according to their functions, as shown in Figure 3:

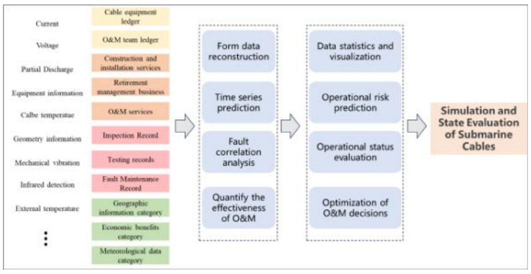

Figure 3. Functional Structure-Based Data Classification Results

Through the research in this section, we have defined the method for data structural classification in this project. Applying this method improves the efficiency of data storage, indexing, querying, and subsequent analysis. This enhances the interpretability of cable condition assessment and lays a solid foundation for conducting effective evaluations of submarine cable status.

2. Data Quality Enhancement

Data quality is the critical factor that affects the models that analyze and assess conditions of submarine cables. The data quality analysis at the early stages of the project reveals several issues, such as sensor failures under certain conditions, incomplete manual inspection records, and data synchronization failures. These problems lead to data gaps that prevent the accurate reflection of the cable equipment's operational state, thereby negatively impacting the reliability of evaluation results. Therefore, it is needed to fill in the missing data with fill-in values which follow the overall pattern of the original data, so that the increase of data quality can be ensured.

2.1 Visualization Analysis of Missing Data

In order to completely understand data gaps, we used analysis tools to visualize data gaps in some key fields, as shown in Figure 4.

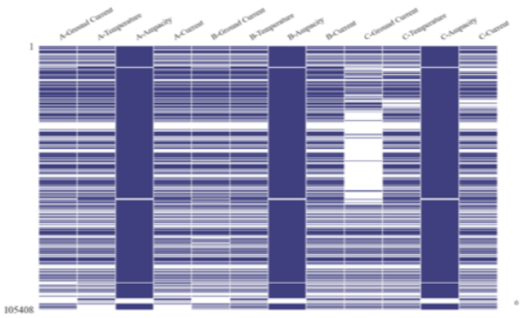

Figure 4. Visualization of Missing Submarine Cable Data

We can see from the figure that the ampacity for Phases A, B, and C of the submarine cable is relatively complete, with minimal missing entries. This is followed by the temperature data and operating current for all three phases. While the grounding current data for Phase A and B also exhibit moderate completeness, the grounding current for Phase C has the most severe missing data issue.

2.2 Statistical Methods for Missing Data Imputation

To make data more intact, we explored several classic statistical methods for missing value imputation by reviewing various sources. The following methods, each with specific advantages, were applied:

Mean Imputation: Suitable for relatively stable datasets.

Median Imputation: More effective for skewed distributions, such as temperature data.

Linear Interpolation: Ideal for restoring the continuity of time-series data like current and voltage.

Time-Series Smoothing Regression: Applicable to trend-based data filling, improves prediction stability through sliding windows and exponential weighted averages.

In the process of filling data gaps, we established upper and lower control limits for each type of data according to their distribution feature and physical meaning, in order to prevent the appearance of unreasonable values.

2.3 AI-Based Methods for Missing Data Imputation

In order to further improve the quality of the filled-out data values, we introduced an AI method that is popular recently -- a Transformer-based architecture -- to predict the missing values.

The Transformer architecture was mentioned by Google in the 2017 paper "Attention is All You Need". It uses Self-Attention structure instead of recurrent structures (e.g. RNNs) commonly used in NLP tasks. Transformers are the foundations of many commonly-seen AI models like ChatGPT, since they enable the model to focus on the most relevant portions of the input.

Essentially, a Transformer follows an encoder-decoder architecture, which means the central Transformer stack is divided into two main components: the encoder and the decoder. The overall structure of the Transformer model is illustrated in Figure 5.

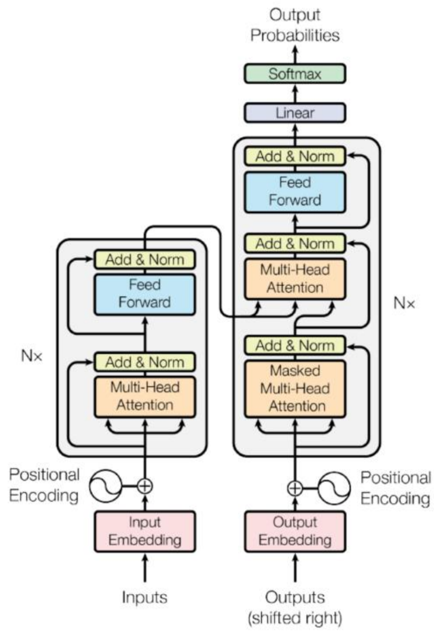

Figure 5. Overall Transformer Architecture

We selected appropriate pretrained model as the base and fine-tuned it using relatively complete data segments from the dataset. Once the model converged during training, it was used to predict and impute the missing values.

2.4 Comparison and Conclusion of Various Imputation Methods

We compared the pre-mentioned methods based on actual datasets, the assessment criteria include the following two values:

1. Mean Absolute Error (MAE): This metric is insensitive to outliers and easy to interpret, making it suitable for tasks where an intuitive understanding of error magnitude is needed.

2. Root Mean Square Error (RMSE): This metric is sensitive to outliers and shares the same unit as the target variable. It is appropriate for tasks where large errors must be penalized more heavily.

We used statistical methods and AI-based methods to impute the missing grounding current values of Phase A. The experimental results are shown in Table 1.

Table 1. Comparison of Data Imputation Methods

Imputation Methods	MAE	RMSE
Mean Imputation	0.010971	0.063899
Median Imputation	0.010022	0.058373
Linear Interpolation	0.01458	0.041412
Time-Series Smoothing Regression	0.007141	0.011327
Transformer	0.006785	0.001450

We can get the following conclusions from Table 1.

The MAE values of the four statistical methods are all under 0.015, RMSE values are all under 0.07. Among the four methods, the most effective one is time-series smoothing regression, followed by linear interpolation and median imputation, while mean imputation has minor accuracy. Overall, although the statistical methods for filling data gaps have errors, their stable performance and relatively low computational demands make them suitable for large-scale datasets where high precision is not critical.

The Transformer-based AI method has the lowest error, with only about 0.006 MAE and about 0.001 RMSE, showing a significant superiority over the statistical methods. However, it requires extensive computational resources, complex setup, and a substantial amount of complete training data. Furthermore, model fine-tuning may result in performance degradation or failure of convergence, demanding significant human and technical resources. It is best suited for tasks that specifically require high accuracy.

Our recommendation: For high-priority data (such as critical cable condition parameters), AI-based imputation is recommended. On the other hand, for secondary data or situations where the cost of AI imputation is prohibitive, statistical methods should serve as a practical alternative.

3. Insights and Reflections

Through continuous involvement in the scientific research project “Research on Simulation and Condition Assessment Technology for Submarine Cables Driven by Data and Physics”, and under the guidance of professors and experts, I have gained significant experience and insights, mainly including the following points.

3.1 Conducting Research Standing on “the Shoulders of Giants”

This research would have been impossible without the long-term efforts and foundational work of State Grid and numerous pioneering researchers. In the process of searching for literary sources, standards, and previous research achievements, I deeply realized that scientific research is a systematic job that requires iteration and extension based on the work of predecessors. Their contributions cleared away a great number of blockages on ours, and gave me a deeper understanding of the phrase “standing on the shoulders of giants”.

In particular, in the construction of the cable condition assessment framework and the design of AI model training, previous research has provided us with valuable theoretical guidance and practical experience. For example, Professor Zhou Chengke’s comprehensive review on power cable asset assessment and operation strategies provided reference for our understanding of the current technical landscape. In addition, large open-source models and datasets from major AI research institutions and teams have laid a solid foundation for our AI model development.

3.2 Artificial Intelligence as a Technological Trendsetter

During data processing, I felt the strength of AI technologies. Particularly, in predicting data loss and constructing models, AI technologies show efficiency and accuracy that surpass traditional statistical methods. In the future, data analysis in power systems will definitely become more dependent on AI models to drive intelligent operation and refined management.

In addition, AI technologies also offer more possibilities for condition sensing and fault warning of submarine cables. For instance, the application of Graph Neural Networks (GNNs) in fault path localization and Long Short-Term Memory (LSTM) networks in predicting temperature trends offer reliable groundwork for future research expansion.

3.3 Future Research Directions

In the next phase, I hope to continue participating in follow-up tasks of this research project. I will put my priority on learning AI modeling technologies, and use these technologies to improve the effectiveness of submarine cable condition assessment. By integrating various structural data, historical fault instances, and expert knowledge, I strive to achieve more accurate diagnosis and trend prediction of cable conditions.

4. Summary

4.1 Summary of Learning

After thorough preliminary preparation and thoughtful guidance from professors and experts, this research project helps me to systematically understand the data types, O&M requirements and methods for analysis in submarine cable engineering. From the collection and structural classification of raw data to the imputation of missing data and the testing of assessment models, this experience not only sharpened my research mindset but also improved my practical skills in programming and data analysis.

Through team collaboration, I obtained the ability to make interdisciplinary communication and advance work collaboratively. I also developed an understanding of the balance between practical engineering and scientific modeling, gradually forming a habit of building technical solutions around real-world problems.

4.2. Improvement and Enhancement

Due to limitations in my experience and time, there is still room for improvement for the structural classification and AI modeling in this project. In the next steps, we hope to include more data of past fault cases, construct more challenging testing datasets, and enhance the model's generalization capabilities. Meanwhile, model deployment and feedback mechanisms should also be strengthened, to create a close-loop optimization process and thereby achieving true synergy between data-driven and physics-based modeling.

Moreover, some aspects of data collection remain incomplete. In the future, we can expand the scope of data collection and integrate it more closely with practical engineering requirements.

Through this experience, I am more certain about my future interest in scientific research, and have deeper insights about research methodologies and technical pathways. I hope to continue growing through future learning and practice, and to explore more meaningful research topics.

Finally, I would like to once again express my sincere gratitude to all the professors and experts who have provided me with patient and thorough guidance throughout this project!

Reference

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